

Jeslin P James

✉ jeslin@ug.cusat.ac.in | 🌐 jeslinpjames.in | 📄 jeslin-p-james | 📞 jeslinpjames | 📞 +91 9383468783

EDUCATION

Cochin University of Science and Technology, Kochi, India Aug 2022 – May 2027 (Expected)
Five Year Integrated M.Sc. in Computer Science (AI & Data Science)

Saint Francis School, Vadakkencherry 2019 & 2021
ISC (Class 12, 2021) — ICSE (Class 10, 2019) – Computer Science

EXPERIENCE

AI & Backend Engineer Intern Aug 2025 – Mar 2026
Laennec AI

- Lead AI research for respiratory sound classification, architecting a CNN+BiLSTM+Attention pipeline in PyTorch; achieved 98% accuracy (up from 86%) via custom feature engineering and K-fold ensembling.
- Own and manage cloud infrastructure on **Google Cloud Platform (GCP)**, configuring GCS buckets and Compute Engine instances for high-performance model training and scalable deployment.
- Develop and maintain robust backends using FastAPI to serve model inferences, ensuring low-latency communication between the AI core and the application layer.

Backend & AI/ML Developer (Contract) Nov 2024 – Apr 2025
Provez, Dubai / Remote

- MVP *Legal Case Manager*: case/matter CRUD, doc ingest/search (PDF, Word, scans), calendar, LLM drafting/summaries; multi-tenant ready.
- Owned backend: FastAPI, Postgres/Alembic, JWT auth, Docker; CI (pytest+GitHub Actions).

Backend & AI/ML Development Intern Aug 2024 – Nov 2024
GreenFI AI (ESG / Carbon Analytics), Singapore / Remote

- ESG emissions pipeline: ingest multilingual utility bills (PDF/img/CSV) → OCR/LLM parse (OpenAI) → normalized energy for carbon calc.
- Asset linking via embedding search (pgvector/Postgres); batch ZIP upload; Flask API on AWS EC2; cut manual asset mapping ~90% on pilot (500 bills).

Student Research Intern May 2024 – Jul 2024
Indian Institute of Space Science & Technology (IIST), Thiruvananthapuram

- Early crop pest detection: built synthetic data pipeline (21K images + 1K real, 11 pest classes) via custom compositing tuned for small objects.
- YOLO variants & FocusDET slice proto; Darknet/CUDA C++ training (~6× faster vs PyTorch proto) and improved small-object detection in field samples.

Open Source Contributions 2024

- 📦 **roboflow/supervision** (computer vision utilities)
- 📦 **hpcaitech/Open-Sora** (video generative AI)

PROJECTS

Multi-Class Object Localization with Deep RL – Developed a GRU-based agent using Grad-CAM and 6 dynamic actions to localize objects across multiple classes.

Statistical Image Synthesis via Matrix Variate Distributions – Developed a framework using pixel-wise mean, variance and neighborhood covariance (matrix variate normal) to directly sample and generate high-fidelity grayscale images; validated on MNIST and Fashion-MNIST with strong SSIM scores.

Share More: P2P File Sharing and Watch Party Platform - Developed a platform for P2P file sharing and real-time watch parties. GitHub

StudySphere: Personalized Learning Platform - Built a platform using React and Django to assist students with notes, flashcards, and quizzes.

Lunar Lander Development with Reinforcement Learning - Created a reinforcement learning model to autonomously land a lunar module.

Kathakali Navarasa Classification Using Vision Transformer (ViT) and LSTM - Developed a model to classify traditional Kathakali expressions from video data.

Implemented Transformer for Malayalam Language Model - Fine-tuned a Transformer model for the Malayalam language.

Custom MLP for Heart Disease Prediction - Built a heart disease prediction model using a Multi-Layer Perceptron implemented with NumPy.

Image Compression - Implemented multiple image compression techniques including DDBTC, MBTC, AMBTC, BTC, and SVD.

TECHNICAL SKILLS

Languages: Python, C++, C, SQL Java, JavaScript, R, LaTeX, Golang

ML/CV/NLP: PyTorch, TensorFlow, OpenCV, CUDA, HuggingFace Transformers

Backend & Data: FastAPI, Flask, Postgres (pgvector), Alembic, Docker, GitHub Actions

Cloud/LLM: AWS (EC2/S3), GCP, OpenAI API